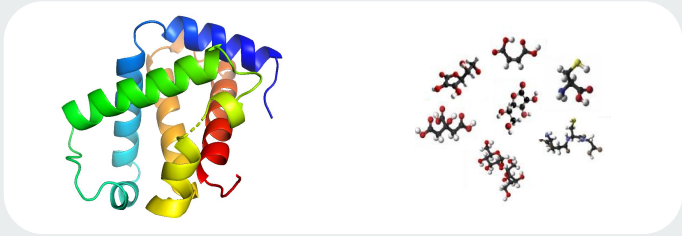
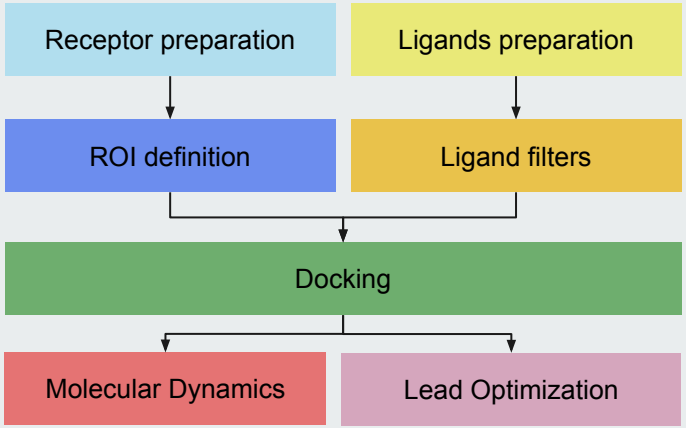
Introduction to virtual drug screening in Scipion-chem

0. Introduction
1. Data acquisition
2. Receptor and ligands preparation
    - 2.1. Regions Of Interest (ROI) definition
    - 2.2. Ligand filters
3. Docking
    - 3.1. Docking analysis
4. Lead optimization (bonus track)
5. Molecular dynamics (bonus track)

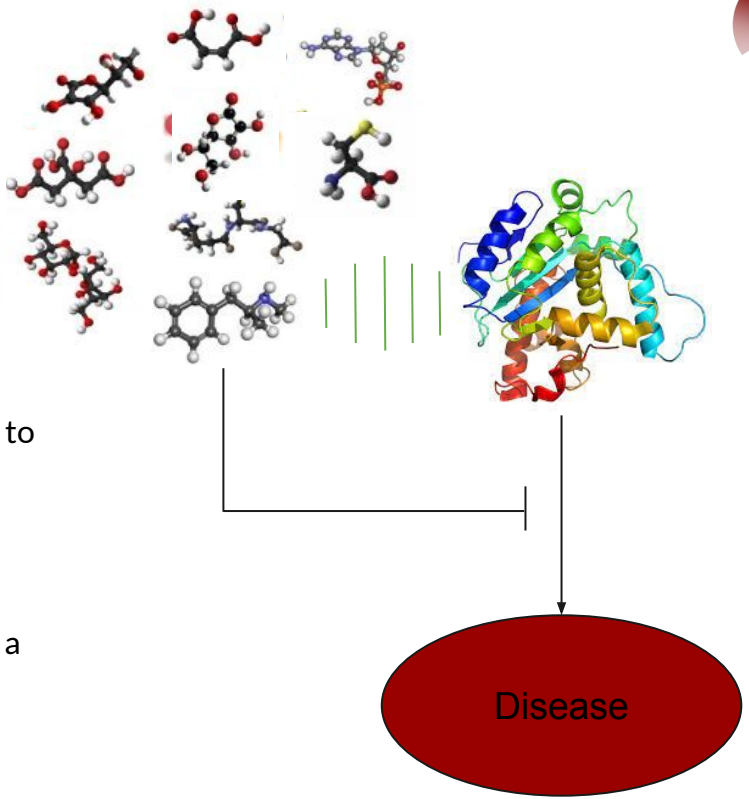




0. Introduction

What is a drug?

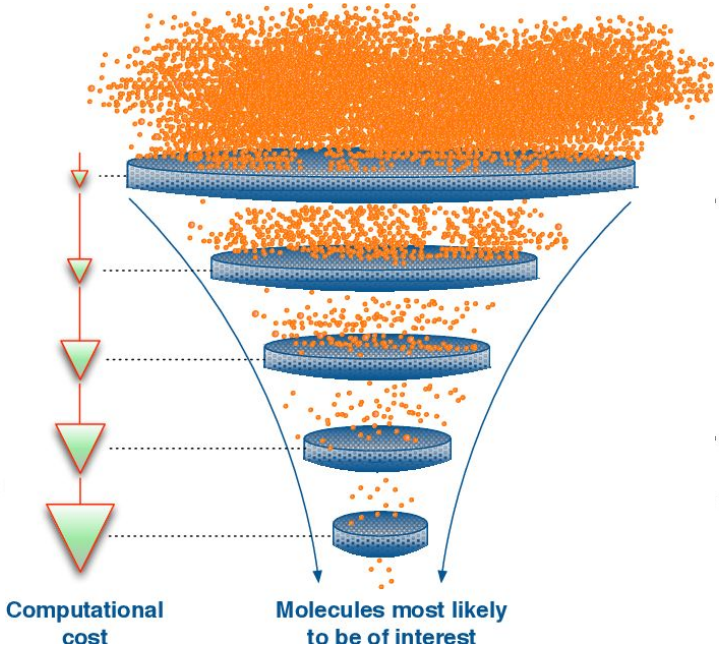
- **Drug:** “Any substance that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. ” **-> Small molecules**
- **Target:** protein involved in a specific biochemical process related to the disease.
- **Mechanism of action:** physico-chemical interaction of the drug with an specific **target** that alters its function
- **Virtual Drug Screening**: In-silico search of **drug candidates** from a **large collection of chemical compounds**



0. Introduction

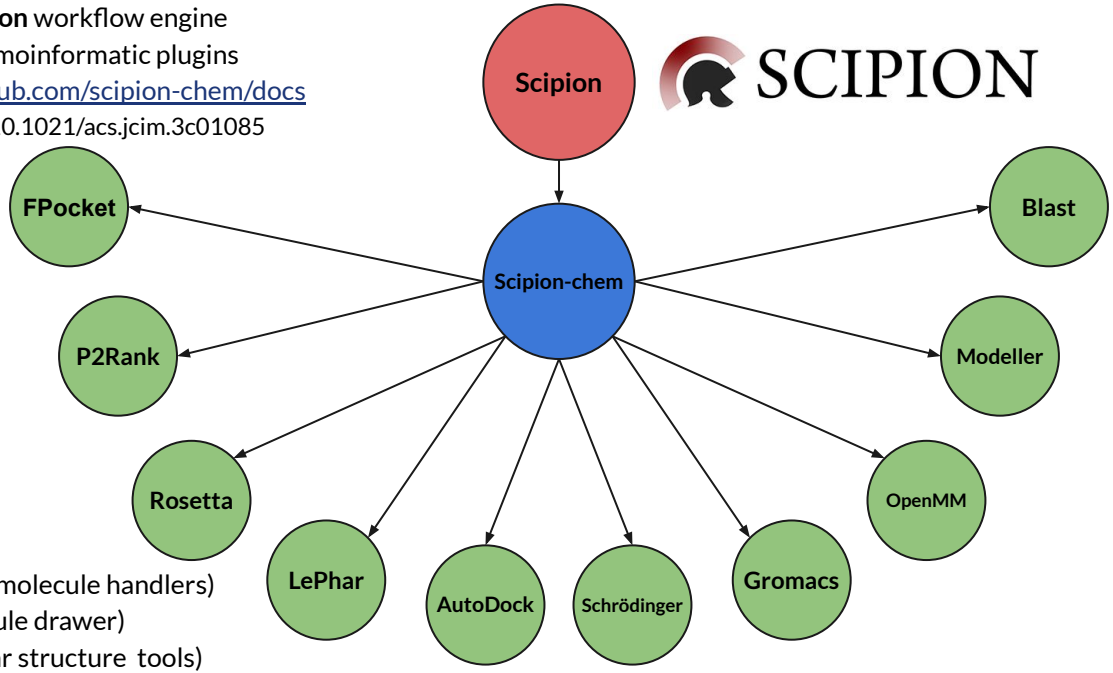
Why using in-silico approaches?

- "**Large collection of chemical compounds**" -> huge **chemical space** of possible drugs ~ 10^{63}
- **In-vitro:** can test even thousands, but it is a **costly process** both in **time** and **resources.**
- **Virtual Drug Screening (VDS):** millions to thousands/hundreds candidates. (But just **predictions**!)
- **Virtual Drug Screening (VDS)**: different steps and approaches developed by **different groups** in **different programs**.
    - Rarely **interoperable** -> Parsing, format conversions, ...
    - Somehow **inaccessible** -> Difficult commands and parameters
    - Difficult to **trace** -> Intermediate files, lost used parameters, ...
    - Hard to **compare** -> Different programs = different results



0. Introduction

- Developed on **Scipion** workflow engine
- New **branch** of chemoinformatic plugins
- **Docs** in https://github.com/scipion-chem/docs
- **Paper** on Dec 2023: 10.1021/acs.jcim.3c01085



Scipion-chem includes:

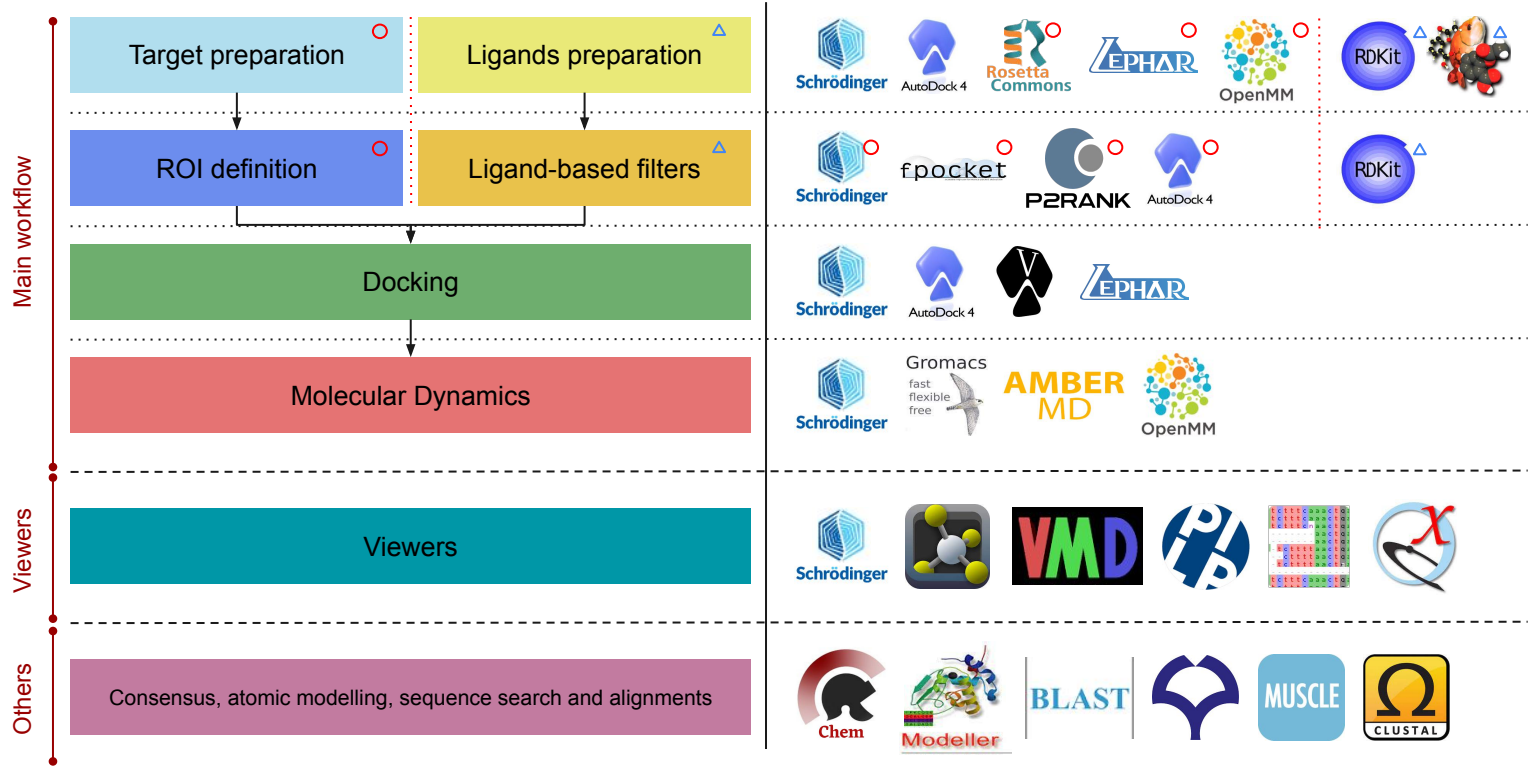
- **Openbabel, RDKit** (molecule handlers)
- **JChemPaint** (molecule drawer)
- **MGLTools** (molecular structure tools)
- **PyMol, VMD** (molecular structure viewer)
- **PLIP** (docking viewer)

15(+) Scipion-chem plugins:

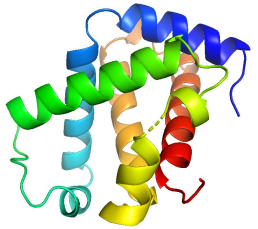
- Pocket prediction
  - FPocket
  - P2Rank
- Docking (and more)
  - AutoDock
  - Rosetta
  - LePhar
  - Schrodinger
- Molecular dynamics
  - Gromacs
  - OpenMM
- Others
  - Modeller
  - BLAST

~ 100 Scipion-chem protocols

0. Introduction

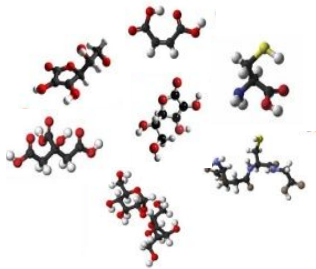


1. Data adquisition



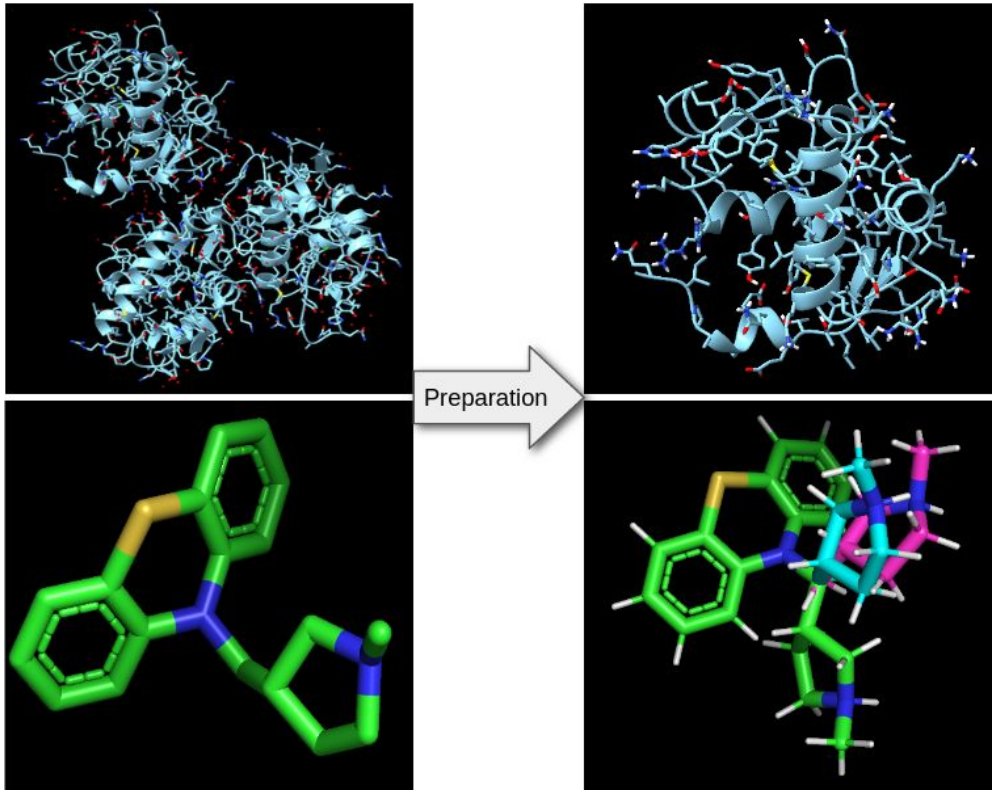
- **Receptor:** usually protein structure
  - Modelling from sequence (Cryo-EM, crystallography, AlphaFold)
  - Download structure PDB or CIF from **PDB database**

- **Ligands:** small molecules libraries
  - **SMI**: string representation of molecules (CCCCCC1=CC(=C2[C@@H]3C=C(CC[C@H]3C(OC2=C1)(C)C)C)O)
  - Molecule structures: **sdf**, mol2, pdb formats
  - Draw molecules (**JChemPaint**)



2. Receptor and ligand preparations

- **Receptor** preparation:
  - Remove **heteroatoms**
  - Remove structure **chains**
  - **Ionization** at a PH
  - Add **hydrogens**
  - Remove **waters**
  - ...
- **Ligand** preparation
  - **Ionization** at a PH
  - Add **hydrogens**
  - Generate **conformers**
  - ...



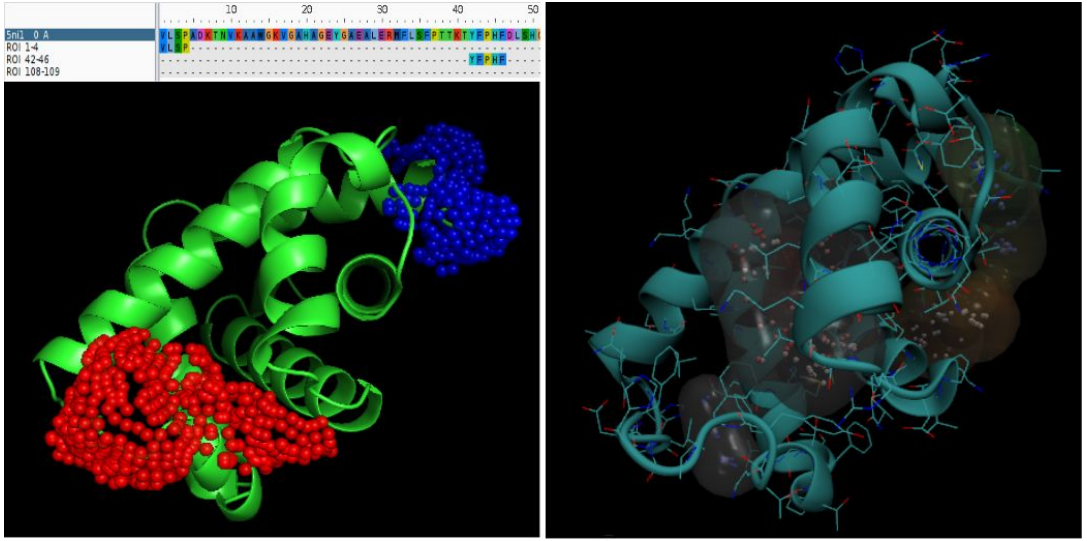
2.1. Regions Of Interest (ROI) definition

Regions of interest (ROIs)

- **Manual** definition:
  - By residue, coordinates, PPI…
- Protein **pockets**:
  - **Potential** ligand locations
- Mapped from **Sequence ROIs**

Consensus ROIs:

- **Comparison** of different source ROIs
- Preserve most **promising** regions
- **Clustering** based on **contact residues**

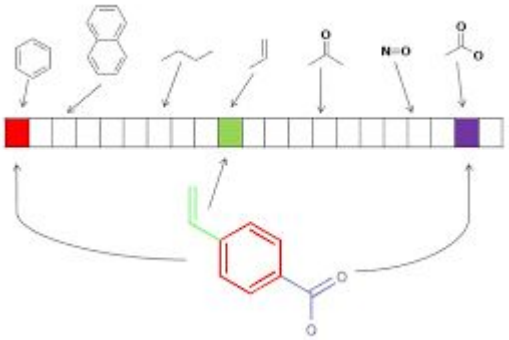


2.2. Ligand filters

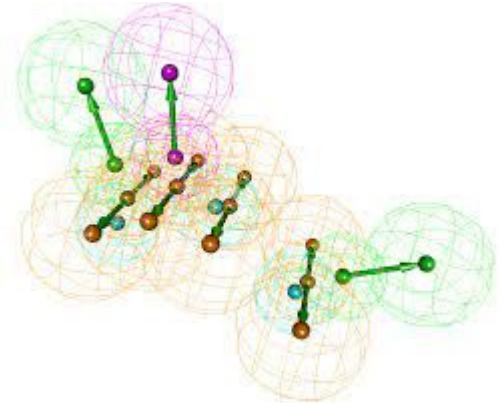
Ligand filters: Take out not promising ligands

- Ligand **chemical** features (**drug-like** ligands):
  - Number of atoms, type of atoms, …
  - **ADME** filter: absorption, distribution, metabolism and excretion
  - **PAINS** filter: highly reactive ligands

- **Similarity** to known ligands:
  - **Shape** similarity: Tanimoto, RMSD…
  - **Fingerprint** similarity: Morgan, MACCS…
  - **Pharmacophore:** 3D **molecular features**



Fingerprints

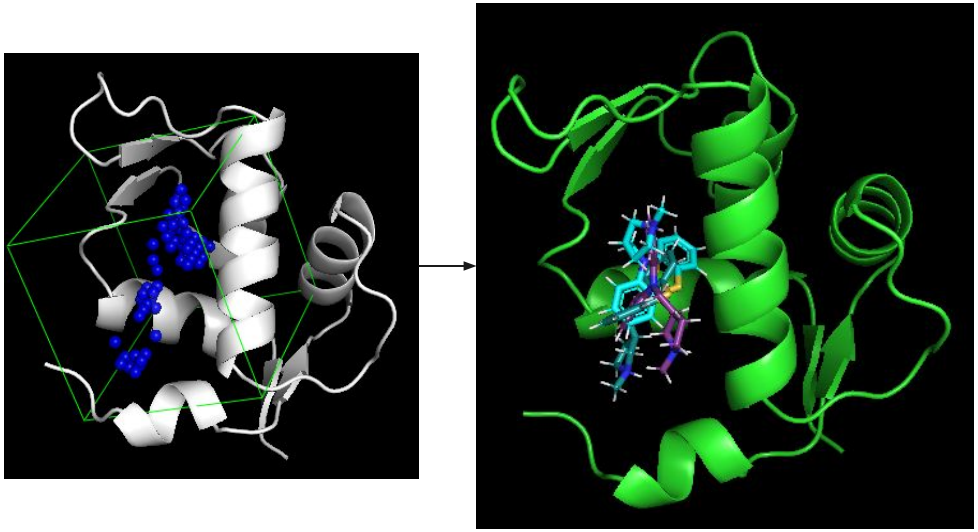


Pharmacophore

3. Docking

Pose ligands on target:

- Matching **shape** and **electrostatics**
- Localized on **ROI** or on the **whole protein**
- **Predicts** receptor-ligand **interaction**



Consensus docking:

- **Comparison** of different software predictions
- **Preserve** most promising docking poses
- **Clustering** based on **RMSD**

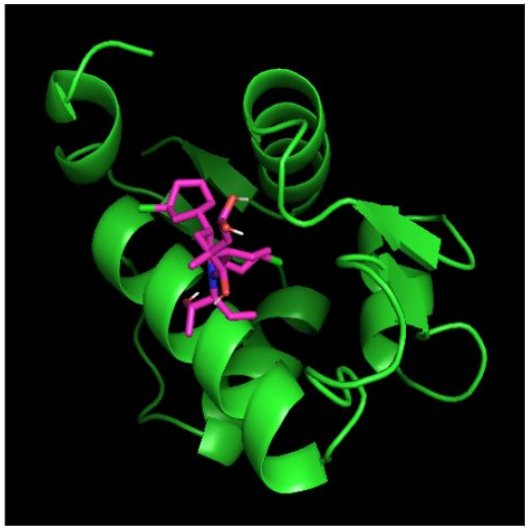
3.1. Docking analysis

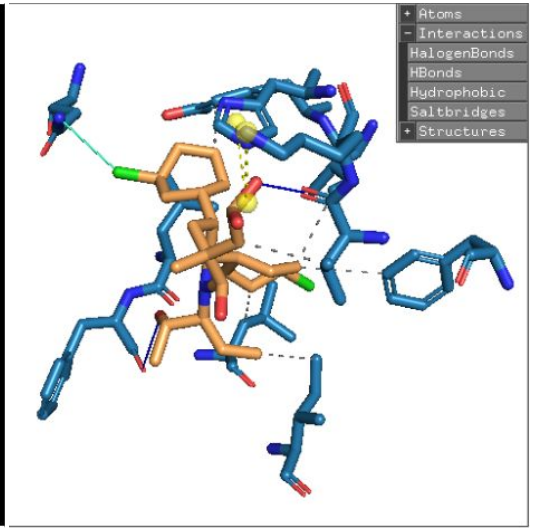
Docking characterization

- **PLIP** (Protein-Ligand Interaction Profiler)**:**
  - Visualization of interactions in pymol

Docking rescoring: External scoring for comparison

- **AutoDock4:** pose rescoring
- **ODDT Scores** (Open Drug Discovery Tool)
  - Vina, RFScore, NNScore and PLECScore
- **Scorch2:** ML model

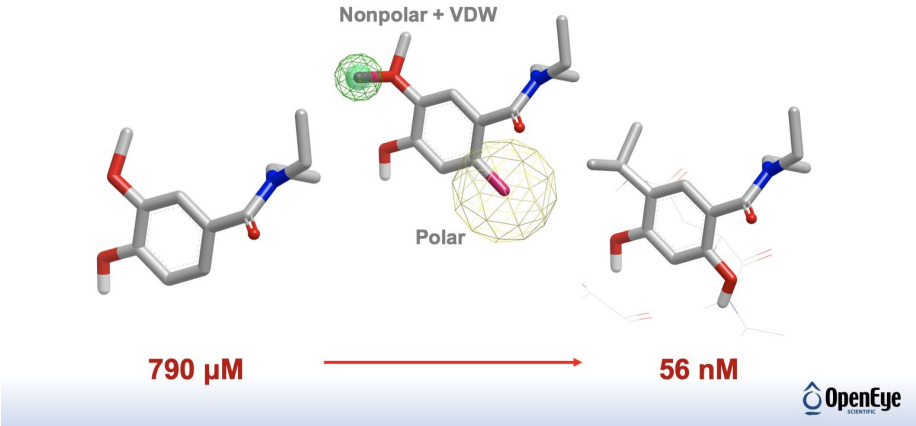




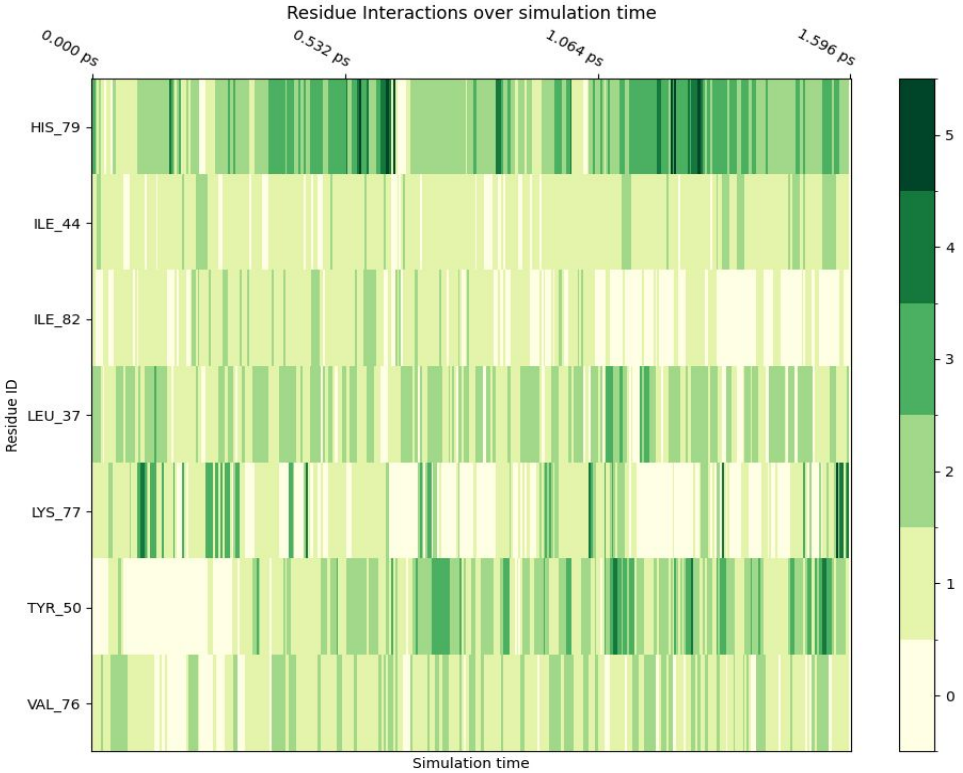
4. Lead optimization (Bonus track)

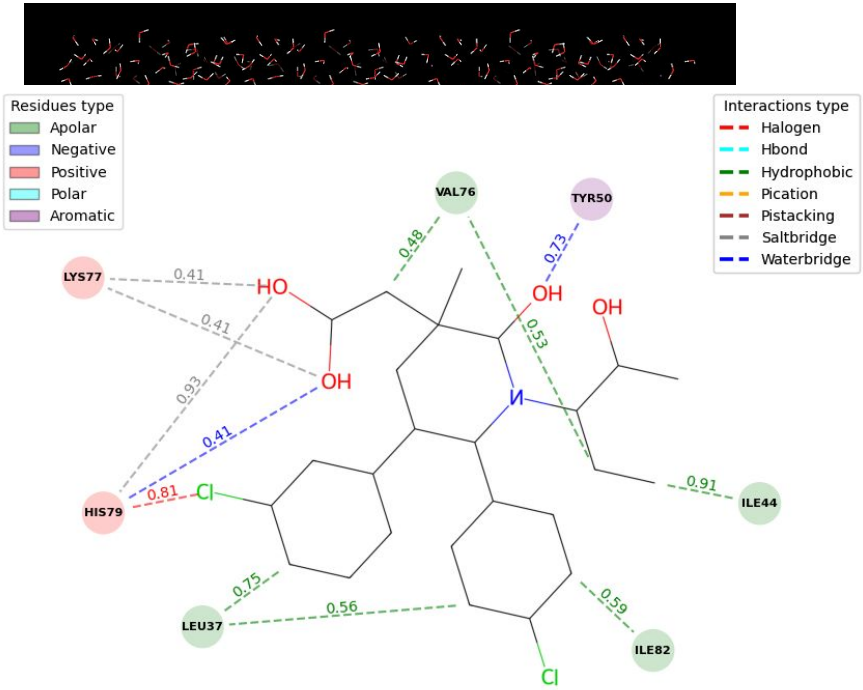
Optimize ligand for interaction

- Ligand **chemical variations**
- Tries to improve:
  - **Affinity**
  - **ADME** properties
- New ligands must be **redocked**



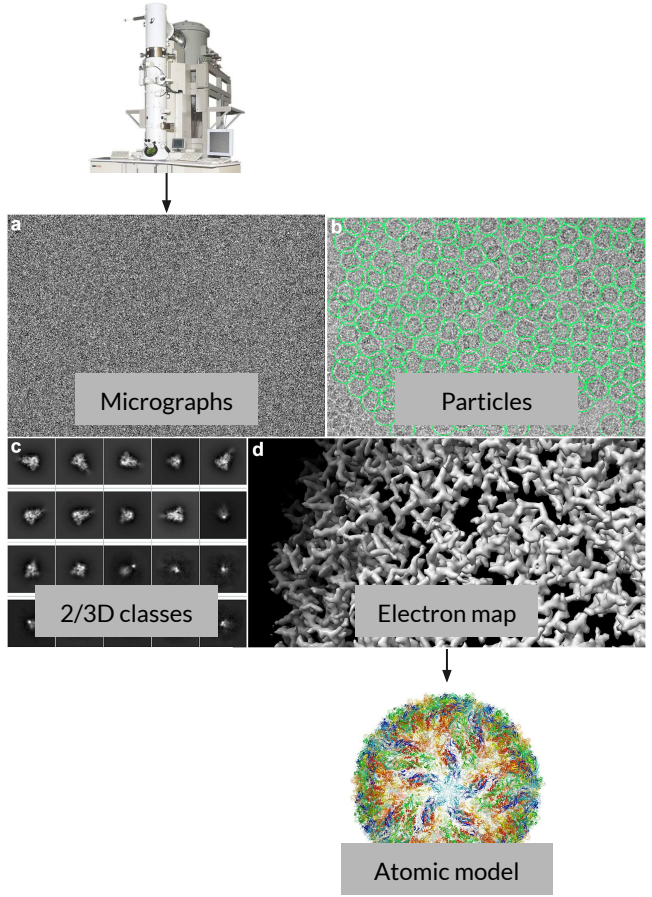
5. Molecular dynamics (Bonus track)





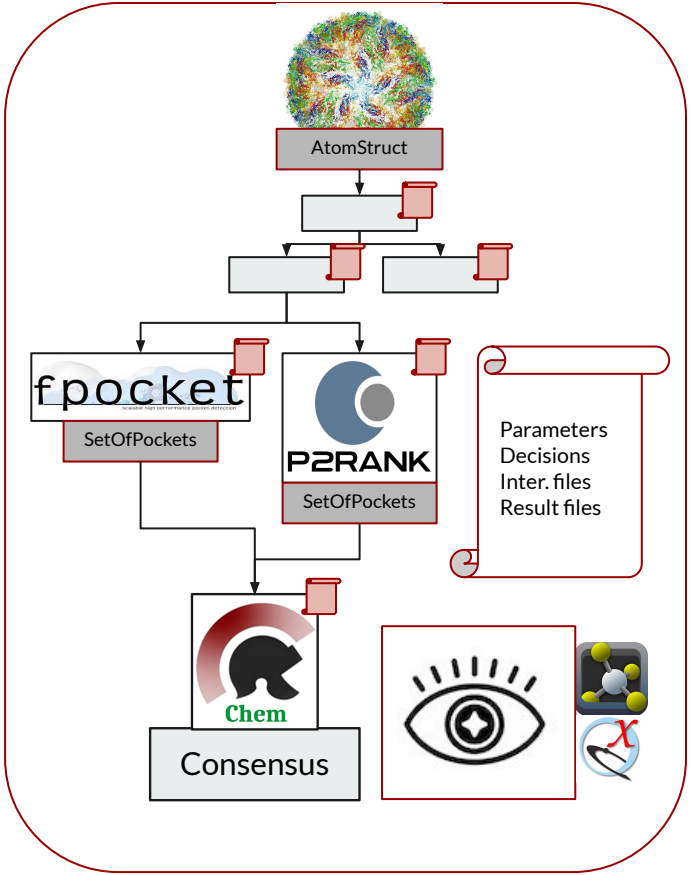
Introduction to Scipion

- **Open platform** that hosts and integrates **bioinformatics software**
- Originally focused on **cryo-electron microscopy**
  - **Single Particle Analysis (SPA)**
  - Tomography
  - Model building

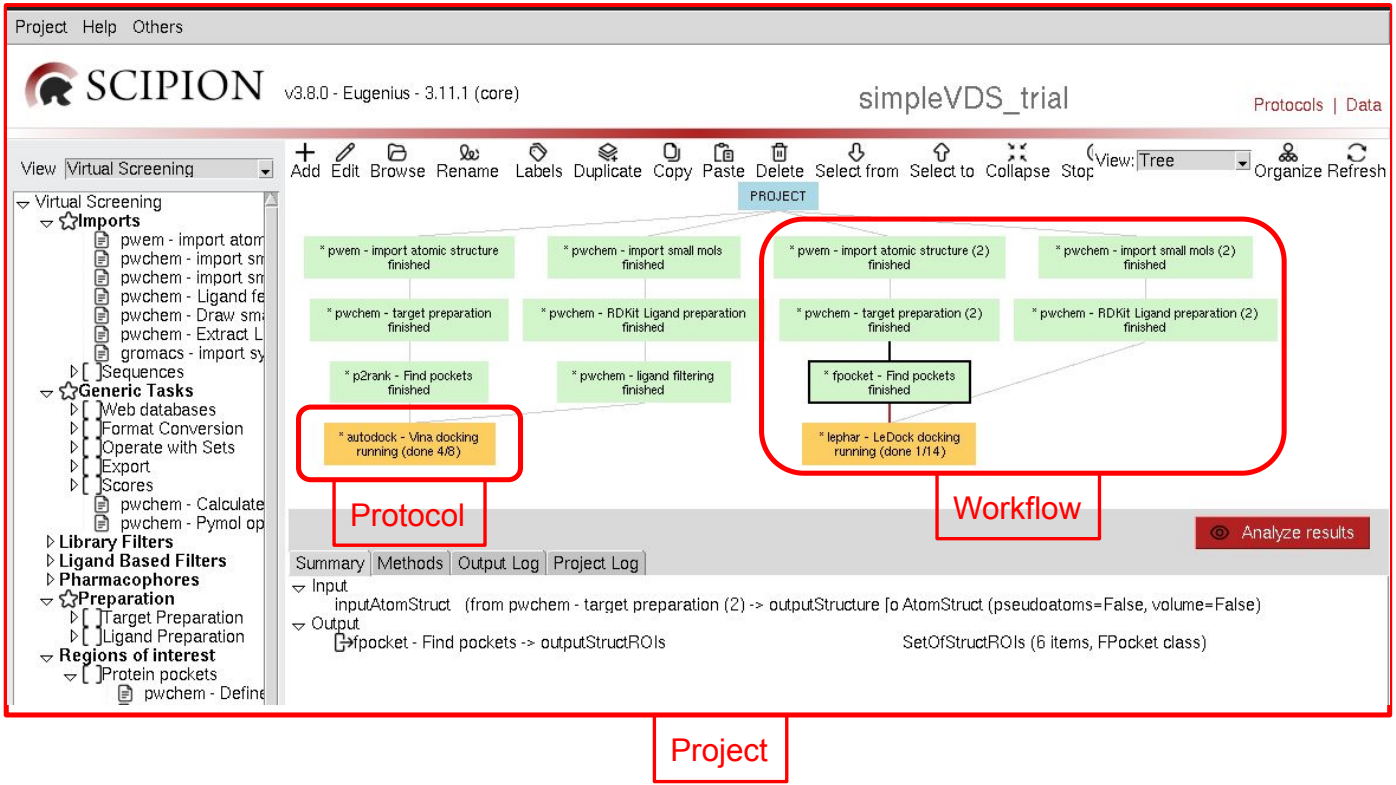


Introduction to Scipion

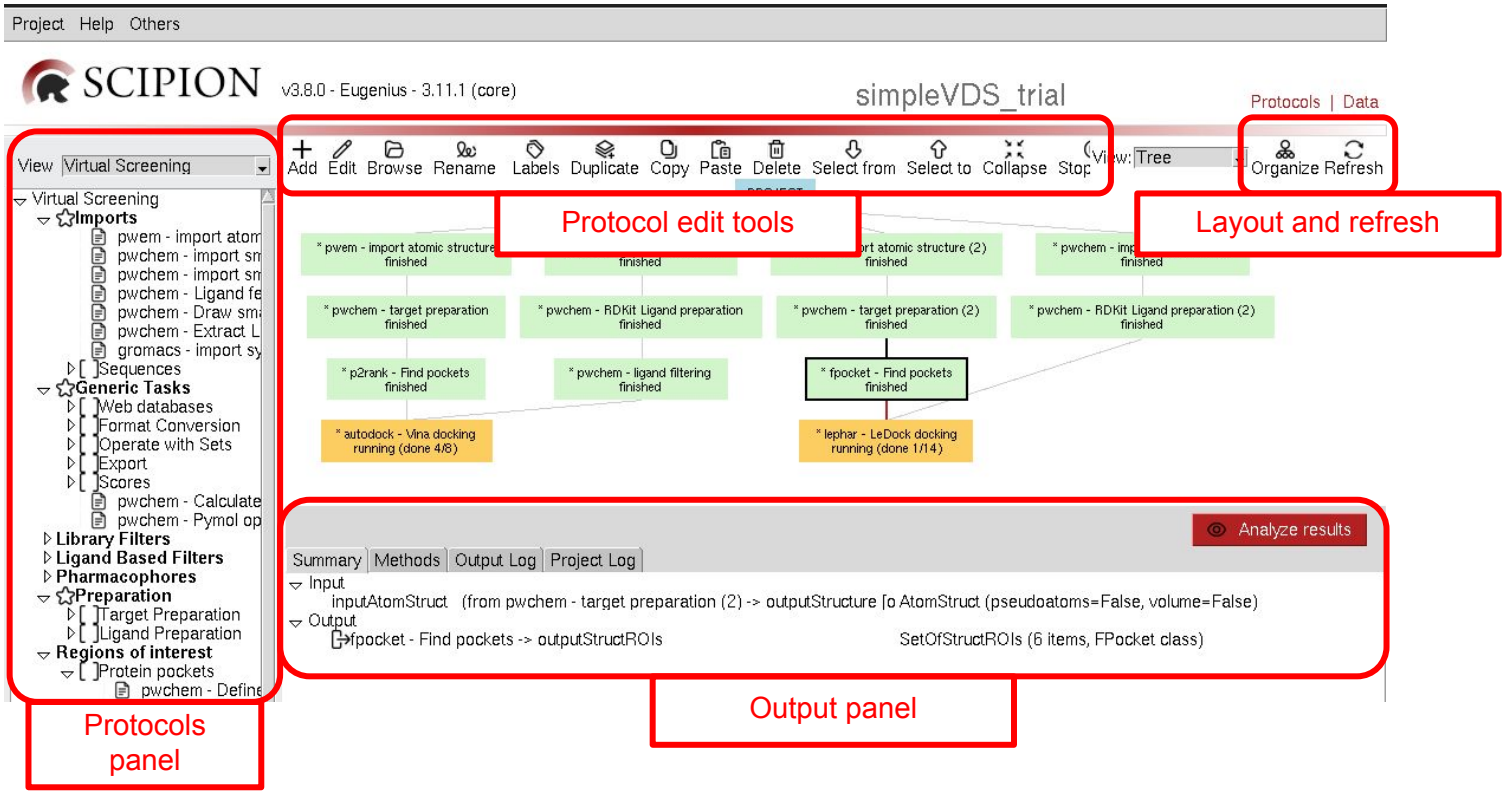
- Manages **workflow executions, software installation** and **interoperability** between programs
- Defines **objects** that represent bioinformatic **entities** and **connect** the different protocols: atomic structure, set of small molecules, pharmacophore…
- Introduces **consensus methods** to **compare** different software results
- **Saves** all input **parameters**, intermediate decisions and files for **traceability**
- Includes **viewers** to analyze the results



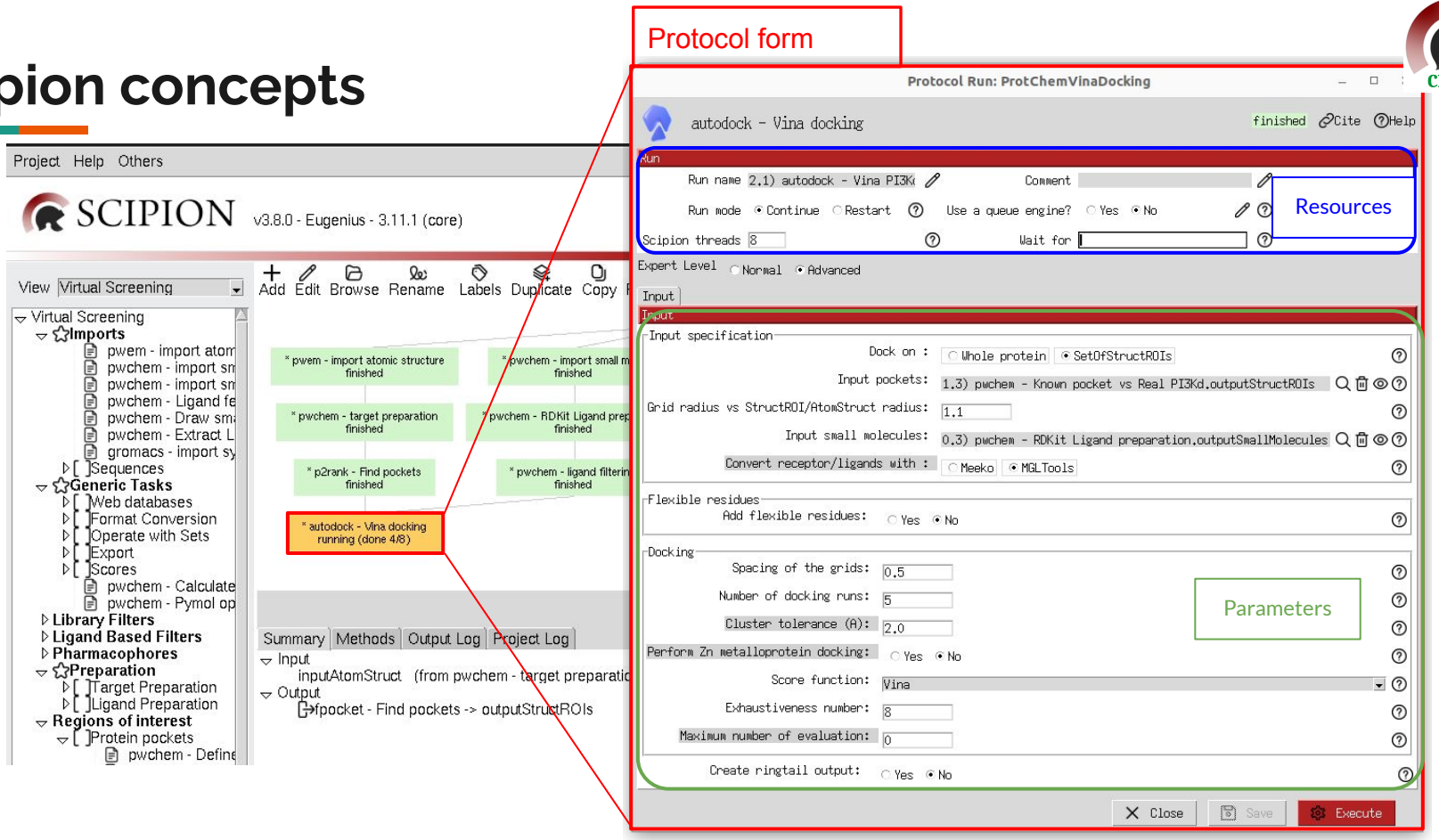
Scipion concepts



Scipion concepts



Scipion concepts



Scipion concepts

